METHODOLOGY

1. DATA COLLECTION

The logistics dataset is collected in CSV format containing shipment details, delivery information, vendor details, and transportation data.

2. DATA PREPROCESSING

The dataset is cleaned by:

- Removing missing values
- Removing duplicate records
- Converting date columns
- Standardizing delivery status values

3. EXPLORATORY DATA ANALYSIS (EDA)

EDA is performed to analyze:

- Shipment mode performance
- Vendor efficiency
- Delivery delays
- Route performance
- Shipping cost distribution

4. FEATURE ENGINEERING

New features are created:

- Delay_Days
- Efficiency_Score

These features improve the prediction capability of the Machine Learning model.

5. MACHINE LEARNING MODEL

Linear Regression algorithm is used to predict shipment delay based on shipping cost and distance.

6. VISUALIZATION

Graphs and charts are generated using Matplotlib and Seaborn to understand logistics performance.

7. DASHBOARD DEVELOPMENT

A Streamlit dashboard is created to display:

- Shipment analysis
- Vendor performance
- Delivery status
- Delay insights

8. RESULT ANALYSIS

The final results help identify inefficient routes, delayed deliveries, and logistics improvement opportunities.